

Assume that the augmented matrix of the system of linear equations is given as follows:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 3 & 2 & -1 & 3 \\ -1 & -1 & 2 & -1 \end{array} \right]$$

Use the following steps to **reduce** the augmented matrix to **generalized row echelon form**:

Step 1:

$$R_3 + R_1 \longrightarrow R_3$$

$$R_2 - 3R_1 \longrightarrow R_2$$

Step 2:

$$-R_2 \longrightarrow R_2$$

☐ $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right]$

☐ $\left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right]$

☐ $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right]$

☐ None of the given options

☒ $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right]$

Given two $n \times n$ invertible matrices A and B such that $(A - B)(B + A) = A^2 - B^2$. Which of the following statement is correct?

- ☐ None of the given options
- ☐ A is the inverses matrix of B
- ☐ $B = BABA^{-1}$
- ☐ $B = ABA^{-1}B$
- ☒ $B = A^{-1}BA$

[Clear my choice](#)

Consider the matrices below with $X = AB$, where

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 3 & -1 & 1 & 3 \\ -2 & 5 & -k & k+1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & 2 & & -1 \\ 0 & 4 & & 1 \\ 3 & -1 & & 1 \\ -2 & 5 & k(k-1) & \end{bmatrix}$$

For which value(s) of k will the entry X_{23} be zero?

- ☐ $k = 3$
- ☐ No such k
- ☐ $k = -2$
- ☒ $k = \frac{1 \pm \sqrt{5}}{2}$
- ☐ $k = 0$ or $k = 1$
- ☐ None of the given option

Clear my choice

Assume that the augmented matrix of the system of linear equations is given as follows:

$$\left[\begin{array}{ccc|c} 2 & 1 & -3 & 2 \\ 2 & -3 & 1 & 10 \\ -2 & 1 & 1 & -6 \end{array} \right]$$

Find the solution of the above system.
Choose the correct option.

- ☐ $x = t + 2, y = 5, z = t$
- ☒ $x = t + 2, y = t - 2, z = t$
- ☐ None of the given options
- ☐ $x = -2t, y = 5 - t, z = -t$
- ☐ $x = -t, y = t - 5, z = -1$

[Clear my choice](#)

Consider the system of linear equations

$$x + y - z = 2$$

$$3x + 2y - z = 3$$

$$-x - y + 2z = -1$$

Write down the system in matrix form,

$$AX = b \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Write down the matrices A and b as well as their respective sizes:

- ☐ $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$, A is a 3 by 3 matrix and b is a 3 by 1 matrix
- ☒ $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$, A is a 3 by 3 matrix and b is a 3 by 1 matrix
- ☐ $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & -2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$, A is a 3 by 3 matrix and b is a 3 by 1 matrix
- ☐ None of the given options are correct.
- ☐ $A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$, A is a 3 by 3 matrix and b is a 3 by 1 matrix

Clear my choice

Consider the system of linear equations

$$x + y - z = 2$$

$$3x + 2y - z = 3$$

$$-x - y + 2z = -1$$

Write down the system in matrix form,

$$AX = b \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Write down the coefficient matrix A of the above system. Determine the inverse A^{-1} of A by applying the matrix inverse algorithm, that is by applying the sequence of elementary row operations in the sense $[A|I_3]$ where I_3 is the 3 by 3 identity matrix.

☐ $A = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & -1 \\ 0 & 1 & 0 & 5 & -1 & -2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right]$

☒ None of the given options are correct.

☐ $A = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & -1 \\ 0 & 1 & 0 & 5 & -1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right]$

☐ $A = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & -1 \\ 0 & 1 & 0 & -5 & -1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right]$

☐ $A = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & 1 \\ 0 & 1 & 0 & 5 & -1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right]$

Consider the system given by the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & k^2 - 2k - 3 & k - 3 \\ 0 & -1 & 1 & 3 \end{array} \right]$$

Determine for which value(s) of k will the system have infinitely many solutions?

- ☒ $k = 3$
- ☐ None of the given options
- ☐ $k = -3$ and $k = -1$
- ☐ $k = -1$
- ☐ $k = -3$ or $k = -1$
- ☐ $k = 3$ and $k = -1$

[Clear my choice](#)

Question **8**

Answer saved

Marked out of 5.00

🚩 Flag question

Let A and B be square matrix of the same size. Which of the following statement is correct?

- ☐ $(A - B)^T$ is symmetric
- ☐ AA^T is skew-symmetric
- ☒ $A - A^T$ is skew-symmetric
- ☐ None of the given options
- ☐ $(A + A^T)^T$ is skew symmetric

[Clear my choice](#)

[Previous page](#)

[Next page](#)

The symmetric equation of a line is given by

$$\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z-4}{1}.$$

Find the parametric form of the line.

- ☐ $x = 1 + 2t, y = -2 + 3t, z = 4 - t$
- ☐ $x = 1 - 2t, y = -2 - 3t, z = 4 - t$
- ☐ $x = 1 + t, y = -2 - t, z = 4 + t$
- ☐ $x = 2t - 1, y = 3t - 2, z = t + 4$
- ☐ $x = 1 + 2t, y = -2 - 3t, z = 4 + t$
- ☐ The correct parametric equation of the line is not listed here
- ☐ $x = 1 - 2t, y = -2 + 3t, z = 4 + t$
- ☐ $x = 1 + 3t, y = -2 - 2t, z = 4 + t$
- ☐ $x = 1 + t, y = -2 - 3t, z = 4 + 2t$
- ☒ $x = 1 + 2t, y = -2 - 3t, z = 4 + t$
- ☐ $x = 1 + 2t, y = -2 + t, z = 4 + 3t$

[Clear my choice](#)

Assume that we are given a unique solution of the system given by

$Ax = V$ where A is the coefficient matrix of the system such Ax is the LHS of the system, with x being the unknown and the RHS is V .

Choose the correct augmented matrix that verifies this statement.

☒
$$\left[\begin{array}{ccc|c} 1 & a & 0 & 4 \\ 1 & 0 & 1 & -2 \\ 0 & 0 & b & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

with $a, b \in \mathbb{R}$ with $b \neq 0$

☐
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 2 \\ 0 & 1 & 0 & -1 & 5 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

☐
$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 4 \\ 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

☐ None of the given options

[Clear my choice](#)

Question **11**

Answer saved

Marked out of 5.00

🚩 Flag question

Given three points $A(1, -1, 1)$, $B(1, 0, 2)$, $C(1, 1, 1)$. Find the side b :

-
- ☐ 5
 - ☐ 3
 - ☒ None of the given options
 - ☐ 4
 - ☐ 2

Clear my choice

Previous page

Next page

Let α be a complex number. Use the Euler form of a complex number to solve for α from the expression

$$1^\alpha = -1$$

-
- ☐ $\alpha = \frac{k+1}{n+1}$ for $k, n \in \mathbb{Z}$ with $n \neq 0$
 - ☐ $\alpha = \frac{k+1}{2n}$ for $k, n \in \mathbb{Z}$ with $n \neq 0$
 - ☐ None of the given options is correct.
 - ☐ $\alpha = \frac{2k}{-n+1}$ for $k, n \in \mathbb{Z}$ with $n \neq 1$
 - ☒ $\alpha = \frac{2k+1}{2n}$ for $k, n \in \mathbb{Z}$ with $n \neq 0$

Clear my choice

Let

$$z_1 = 1 - 2i - i^2$$

and

$$z_2 = x + y + ix$$

be two complex numbers such that

$$z_1 = z_2.$$

The quotient $\frac{y-2}{x+y}$ is given by

-
- ☒ 1
 - ☐ -2
 - ☐ 2
 - ☐ -3
 - ☐ -6
 - ☐ $\frac{1}{2}$
 - ☐ 1
 - ☐ None of the given options

Clear my choice

What are the cube roots of $8i$?

- ☐ None of the given options
- ☐ $z_0 = 2 \left[\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right], z_1 = 2 \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right], z_2 = 2 \left[\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right]$
- ☐ $z_0 = 2 \left[\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right], z_1 = 2 \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right], z_2 = 2 \left[\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right]$
- ☐ $z_0 = 2 \left[\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right], z_1 = 2 \left[\cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} \right], z_2 = 2 \left[\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right]$
- ☒ $z_0 = 2 \left[\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right], z_1 = 2 \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right], z_2 = 2 \left[\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right]$

Clear my choice

Find the general form of the matrix that commute with the matrix $\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$

- ☐ $\begin{bmatrix} a & a-d \\ 0 & 2d \end{bmatrix}$ whenever $a \neq d$
- ☐ $\begin{bmatrix} a & a-d \\ 0 & 2d \end{bmatrix}$ whenever $a = d$
- ☒ $\begin{bmatrix} a & a-d \\ 0 & 2d \end{bmatrix}$ whenever $a = 2d$

[Clear my choice](#)

Assume that $C = A - I_3$ where

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -2 & 0 \\ -1 & 0 & 0 \end{bmatrix}.$$

Find the matrix B such that

$$CB^T = 3 \begin{bmatrix} -1 & 2 \\ 3 & 1 \\ 0 & -1 \end{bmatrix}$$

where I_3 is the 3 by 3 identity matrix.

- ☐ $B = \begin{bmatrix} -3 & 4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$
- ☐ $B = \begin{bmatrix} -3 & -4 & 3 \\ 9 & 2 & 6 \end{bmatrix}$
- ☐ $B = \begin{bmatrix} -3 & -4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$
- ☐ None of the given options
- ☒ $B = \begin{bmatrix} 3 & -4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$

[Clear my choice](#)

We consider the system given by the following augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & \alpha^2 - \alpha - 2 & \alpha + 1 \end{array} \right]$$

Determine the value (s) of α for which the system has infinitely many solutions.

-
- ☐ $\alpha = -2$
☐ $\alpha = 1$
☐ $\alpha = 2$
☐ None of the given options are correct
☒ $\alpha = -1$

[Clear my choice](#)

Suppose that

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 3 & 4 & -1 \end{bmatrix} \text{ such that } |A| = 1. \text{ Suppose that}$$

$$D = \begin{bmatrix} 4 & 2 & 6 \\ 0 & -2 & 4 \\ 6 & 8 & -2 \end{bmatrix}$$

Write down $|D|$, the determinant of D .

- ☐ 4
- ☐ -8
- ☒ 8
- ☐ None of the given options are correct
- ☐ -4

[Clear my choice](#)

Identify whether or not the matrix given below is an elementary matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -2 & 0 \end{bmatrix}$$

Note that F stands for False and T stands for True in the below choices. Choose the correct answer.

-
- ☐ T or F
- ☐ None of the given options are correct
- ☐ T
- ☒ F
- ☐ T and T

[Clear my choice](#)